Aaryan S. Maralihalli

AI Product Engineer

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Summary

Engineer who ships **user-facing AI and real-time systems end-to-end** — LLM features, sub-100ms retrieval, event-driven pipelines, and the React/TypeScript front-ends on top. At HyperVerge, owned integrations for **6 consumer-fintech apps** and lifted onboarding **conversion by 10% (to >95%)** by instrumenting drop-off and tuning flows — the same engagement-and-conversion problem Plotline solves. **IEEE-published** (interpretable cardiac ML). CS grad, Jul 2026.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026
AI identity-verification platform for consumer fintech apps

- Owned end-to-end integration for **6 enterprise fintech clients** (HomeCredit, PaytmMoney, Mobikwik, IndMoney+) across **15+ lines of business**; took **3 live at >95% verification conversion — a 10% lift**.
- Drove **post-go-live conversion/drop-off analysis** and stabilization — diagnosed user-journey drop-offs (e.g., IndMoney) and shipped flow optimizations; proactively added modules clients hadn't scoped.
- Built **"HV Deals Bot"** (Python + Slack): tracks every deal's stage and next steps across the team — internal tooling adopted org-wide for process visibility.

Integrated Genetic Solutions, Bangalore
Project Intern, Medical AI & Web Jun 2024 – Aug 2024

- Led migration of a legacy Bootstrap-JS dashboard to **React (MERN)**, improving processing of **5M+ lab values** on a HIPAA-compliant product.

Selected Projects

LANGTRON — AI memory assistant github.com/Aaryansm11/LANGTRON

- Fine-tuned **Mistral-7B (LoRA)**; dual-model architecture; **FAISS semantic retrieval at sub-100ms** — latency engineering for interactive AI UX.

Live Emostream — real-time engagement pipeline github.com/Aaryansm11/Live_emostream

- Event-driven **Kafka + Spark + Redis** with Flask-SocketIO broadcast and nginx load balancing; **load-tested to 1,000 concurrent users** (Locust) — real-time triggers at user scale.

AlgoScope (live)

- Interactive in-browser algorithm visualizer — **React/Next.js + Pyodide** code tracing with animated data-structure views; shipped and deployed (algoscope-beta.vercel.app).

SDIS — secure RAG service github.com/Aaryansm11/SDIS

- FastAPI retrieval service with JWT/RBAC, PII redaction (spaCy), FAISS over PostgreSQL; Dockerized, **pytest + GitHub Actions CI**.

Self-Supervised Cardiac Model (IEEE-published) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- Contrastive learning (SimCLR) on MIMIC-III ECG/PPG; **ROC-AUC 0.757**.

Skills

- **AI/LLM:** PyTorch, HuggingFace, LoRA fine-tuning, RAG, FAISS, embeddings, LangChain, prompt design
- **Real-time / Backend:** Kafka, Redis, Spark, FastAPI, Flask, PostgreSQL, REST, JWT, Docker, AWS, nginx
- **Product / Front-end:** TypeScript, React, Next.js, Framer Motion, analytics-driven iteration (conversion/drop-off), SDK-style client integration; DSA (150+ LeetCode)

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."*
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026